Hand-Object Interaction as an Intermediate Representation for Haptics in GenAI-Driven XR Authoring

DOHUI LEE, Graduate School of Metaverse, KAIST, Republic of Korea

Generative AI is rapidly expanding what can be authored in XR, including scenes, behaviors, and multimodal effects. Yet haptics is still rarely integrated into the same workflow, and is often added afterward through manual mappings to low-level events. We argue that a key missing link between interaction authoring and haptics authoring is an intermediate representation that captures part-level interaction structure in a form usable for feedback generation and binding. We propose Hand–Object Interaction (HOI) as one promising basis, because HOI can represent how users acquire parts, manipulate them under constraints, and produce state changes that are meaningful for haptic feedback. From this perspective, we identify three requirements for using HOI as an intermediate representation in authoring workflows: representing interaction in a haptics-usable form, guiding feedback generation and binding, and supporting designer inspection and revision of inferred mappings. We also discuss implications for the scope of representation, the role of automation, and the evaluation of HOI-based haptic authoring in GenAI-driven XR workflows.

Additional Key Words and Phrases: Generative AI, Haptics, Hand-Object Interaction, Extended Reality, GenAI-Driven Authoring

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1 Introduction

Generative AI is rapidly reshaping XR creation workflows by accelerating how designers author diverse modalities and components of an experience. Recent systems increasingly support generating 3D assets and scenes from high-level intent [1, 2], as well as authoring interactive behaviors and logic [4, 5]. Beyond visuals and interaction, research is also moving toward generating other modalities that shape XR experiences, such as sound [11, 12] and haptics [6, 13].

Despite this rapid progress, today's workflows remain modular rather than integrated. In practice, when haptics is added to generated interactions, designers often still need to inspect event logic, interpret the interaction flow, and manually decide where feedback should be attached [3, 10]. This makes haptic authoring labor-intensive under iteration, and has motivated dedicated authoring toolkits for designing haptic feedback across contexts [7, 9, 14].

We see Hand–Object Interaction (HOI) as an intermediate representation that connects GenAI-driven interaction authoring with haptics generation and binding through a natural structure for feedback design. Building on this view, we explore how HOI can support haptics generation and binding, and outline the challenges that remain.

2 Problem Framing and Position Statement

We ask what it would take to integrate haptics into GenAI-driven XR authoring. The key problem is that current workflows still lack a shared representation between interaction authoring and haptics authoring, leaving feedback generation and binding to be manually inferred from event logic. We therefore explore part-level HOI as an intermediate representation between interaction authoring and haptics authoring. As an initial step, we draw on our experience building HOICraft [8], an AI-assisted in-situ VR tool that makes part-level HOI mappings explicit. We then identify

Author's Contact Information: Dohui Lee, Graduate School of Metaverse, KAIST, Daejeon, Republic of Korea, dohui.lee@kaist.ac.kr.

three design requirements for an interaction–haptics interface and illustrate these requirements through a simple XR authoring scenario.

R1 – Haptics-oriented HOI representation. To support haptics generation and binding, HOI should represent interaction in a form that is directly usable for feedback authoring. At minimum, this includes (1) which interaction moments may call for feedback (e.g., acquisition, contact, release, state transition), (2) which interaction properties matter for haptic design (e.g., manipulated part, action type, constraint, control progression), and (3) what role the feedback is intended to play (e.g., confirmation, guidance, realism, playfulness). The goal is to provide a compact interaction representation that haptics modules can interpret and use.

R2 – HOI-guided generation and binding. Given a haptics-oriented HOI representation, the interface should support using HOI to decide what level of feedback fidelity is appropriate for each interaction moment and how that feedback should be bound to relevant interaction elements. This decision should be guided in part by what role the feedback is intended to play at each moment. For example, confirmation or guidance may be served by lightweight cues, whereas realism-oriented moments may call for richer feedback, such as audio-conditioned generation or physically simulated responses. Therefore, HOI should help determine appropriate fidelity based on feedback role and provide meaningful anchors for cue attachment.

R3 – Inspectability and editability. Because GenAI-based workflows may infer HOI and propose haptic bindings automatically, designers must be able to inspect and revise both. The interface should make the relation between the inferred HOI structure and generated haptic output transparent, so designers can understand why a cue was attached, which interaction element triggered it, and how it may change under interaction edits. It should also support direct refinement of both the underlying HOI representation and the resulting bindings (e.g., removing a cue, changing its timing, or revising its intended role), so designers remain in control of the final haptic experience.

Illustrative Scenario. A designer is authoring a VR coffee machine with interactive parts such as a power button, a rotary dial, and a lever. The interaction authoring system first produces a part-level HOI description for each component. For the dial, it may indicate grasp-and-turn input with discrete progression and identifiable step transitions. For the lever, it may indicate a pull motion with a constrained range and a terminal stop.

Based on this HOI representation, the system proposes haptic mappings for feedback-relevant interaction moments. Lever resistance or end-stop contact may call for highly realistic feedback, while button presses or dial step transitions may be better served by lightweight confirmation cues. Dial step transitions may also use more stylized cues when the goal is to emphasize discrete progression rather than physical realism. The same HOI structure also provides anchors for binding each cue to the appropriate interaction moment.

The designer then inspects the proposed mappings and revises them as needed. They may strengthen the lever's terminal cue for emphasis, simplify the dial feedback to reduce clutter, or disable nonessential confirmation cues. In this way, HOI supports haptics generation and binding while still allowing the designer to shape the final experience.

3 Implications and Future Directions

A key challenge in treating HOI as an intermediate interface for haptics is deciding what information it should encode. If the representation is too sparse, it collapses back into low-level event triggers and cannot support meaningful feedback generation or binding. If it is too detailed, it begins to resemble a full interaction or physics model, making it difficult to reuse across tools and authoring contexts. A practical next step is therefore to define what aspects of HOI should remain stable and authorable and what should remain the responsibility of runtime systems, such as sensing, tracking, and physical simulation.

A second challenge is determining how HOI should be used in automation. HOI may help a system infer what kind of feedback is appropriate, what level of fidelity it deserves, and where it should be bound. However, not all of these decisions should necessarily be automated to the same degree. Designers may welcome automatic suggestions for candidate cues and binding anchors, while still wanting explicit control. An important research question is therefore how to distribute control between semantic inference, automatic generation, and designer revision in ways that support trustworthy co-creation.

Finally, progress will require datasets and evaluation methods that reflect real authoring goals. We need paired examples that connect part-level HOI descriptions with haptic outcomes, binding decisions, and the authoring intent behind them. We also need evaluation criteria that go beyond signal realism to test whether generated haptics supports the intended interaction experience—for example, whether it improves confirmation, conveys resistance, clarifies control progression, or reinforces stylistic tone. A practical next step is to define a small set of reusable HOI primitives and build shared HOI–haptics benchmarks around them, enabling future systems to be trained, compared, and iterated more systematically.

4 Conclusion

We argue that a key missing link between GenAI-based interaction authoring and haptics authoring is an intermediate representation that captures part-level interaction structure in a form usable for feedback generation and binding. We position HOI as one concrete basis for such an interface, and outline how it can represent interaction for haptics authoring, guide feedback generation and binding, and support designer inspection and revision of inferred mappings. We hope this position can help move haptics integration beyond trigger-level attachment toward more structured and reusable interaction–haptics authoring workflows, and we look forward to discussing these ideas with other workshop participants and identifying concrete next steps toward integrated interaction–haptics authoring pipelines.

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